

VEDANSH KUMAR

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New Delhi, India

EDUCATION

Bharati Vidyapeeth's College of Engineering (BVP)

B.Tech in Electronics and Communication Engineering

Aug 2024 – July 2028

New Delhi, India

Adarsh Public School

Senior Secondary – CBSE

Graduated March 2024

New Delhi, India

PROJECTS

AI Drone Detection System | Python, YOLOv8, TensorFlow, OpenCV, Raspberry Pi

2025 – Present

- Building real-time drone detection using YOLOv8 on Raspberry Pi edge hardware; custom labeled dataset targeting 90%+ accuracy across varied lighting for airspace monitoring.
- Deploying model locally on Pi to eliminate cloud dependency and achieve low-latency field inference without internet.
- Exploring integration with alert systems to notify operators upon drone intrusion detection in restricted zones.

DIY Quadcopter Drone | C++, Arduino, Betaflight, JavaScript, HTML/CSS

2024

- Hand-built fully functional quadcopter – frame assembly, ESC calibration, Betaflight PID tuning; achieved stable medium-range hover entirely from scratch.
- Built real-time browser telemetry dashboard (HTML/CSS/JS) showing live battery, altitude and orientation data.
- Documented full build process for reproducibility; planned next phase includes GPS-based autonomous waypoint navigation.

Smart Glove for Speech-Disabled People | Python, ML, Raspberry Pi 4, Java

2024

- Wearable glove using flex sensors + IMU on Raspberry Pi 4; sklearn classifier recognises 20+ gestures with 95%+ accuracy, outputting speech via LCD and speaker in real time.
- Web interface lets users customise gesture-to-phrase mappings for personalised communication.
- Designed system to run fully offline, making it accessible in low-connectivity environments for daily use.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, HTML/CSS, Java, SQL

Tools & Frameworks: TensorFlow, OpenCV, YOLOv8, Arduino IDE, Betaflight, Git/GitHub, VS Code

Hardware: Raspberry Pi 4, Arduino, Flex Sensors, IMU (MPU-6050), ESCs, Flight Controllers

Interests: Computer Vision, Drone Technology, Embedded Systems, Assistive Technology, AI/ML

ACHIEVEMENTS

- **Scholar's Trophy Holder** – Adarsh Public School (2016–2024): academic excellence across 8 consecutive years.
- **Head Prefect of Student Council** – Adarsh Public School (2020–2024): led 30+ volunteers, managed all school-wide events.
- Presented hardware and AI projects at national hackathons including **BrainWave** and multiple inter-college tech fests.
- Represented school badminton team – **District-Level Finals (2023)**; participated in College Judo at inter-college and State level.

LEADERSHIP & VOLUNTEERING

MLSAC – Technical Society, BVP (Public Representative Head)

Sep 2024 – Present

BVP, New Delhi

- Organise workshops, coding contests and speaker sessions for 200+ students; liaison between society and college administration.
- Coordinate technical events and manage member engagement across the society's annual calendar.

Venuva – Modelling Society, BVP (Corporate Executive)

Sep 2024 – Present

BVP, New Delhi

- Manage corporate outreach and sponsorships; coordinate with external brands for event funding and collaborations.

EngiFest – Cultural Society, BVP (Corporate Executive)

Oct 2024 – Present

BVP, New Delhi

- Drive corporate partnerships for BVP's flagship annual tech-cultural fest supporting 1000+ attendees.